Lecture 7: Sep 17, 2026

Last time:

- Transverse vs longitudinal waves
- Travelling waves and the wave equation
- Total phase for sinusoidal waves
- Superposition of waves

Today's class:

- Complex representation for wave addition (interference)
- Waves on a stretched string
- Using dimensional analysis

COMPLEX REPRESENTATION OF A WAVE

Top Hat Question

What is the complex amplitude for the wave $s_1(x,t) = 2.00\cos(kx - \omega t + 60^o)$?

A. $\tilde{s_1} = 2.00e^{j\pi/3}$

B. $\tilde{s_1} = 2.00e^{j60^o}$

C. $\tilde{s_1} = 2.00$

D. $\tilde{s_1} = 2.00e^{j\pi/3}e^{j(kx-\omega t)}$

Example

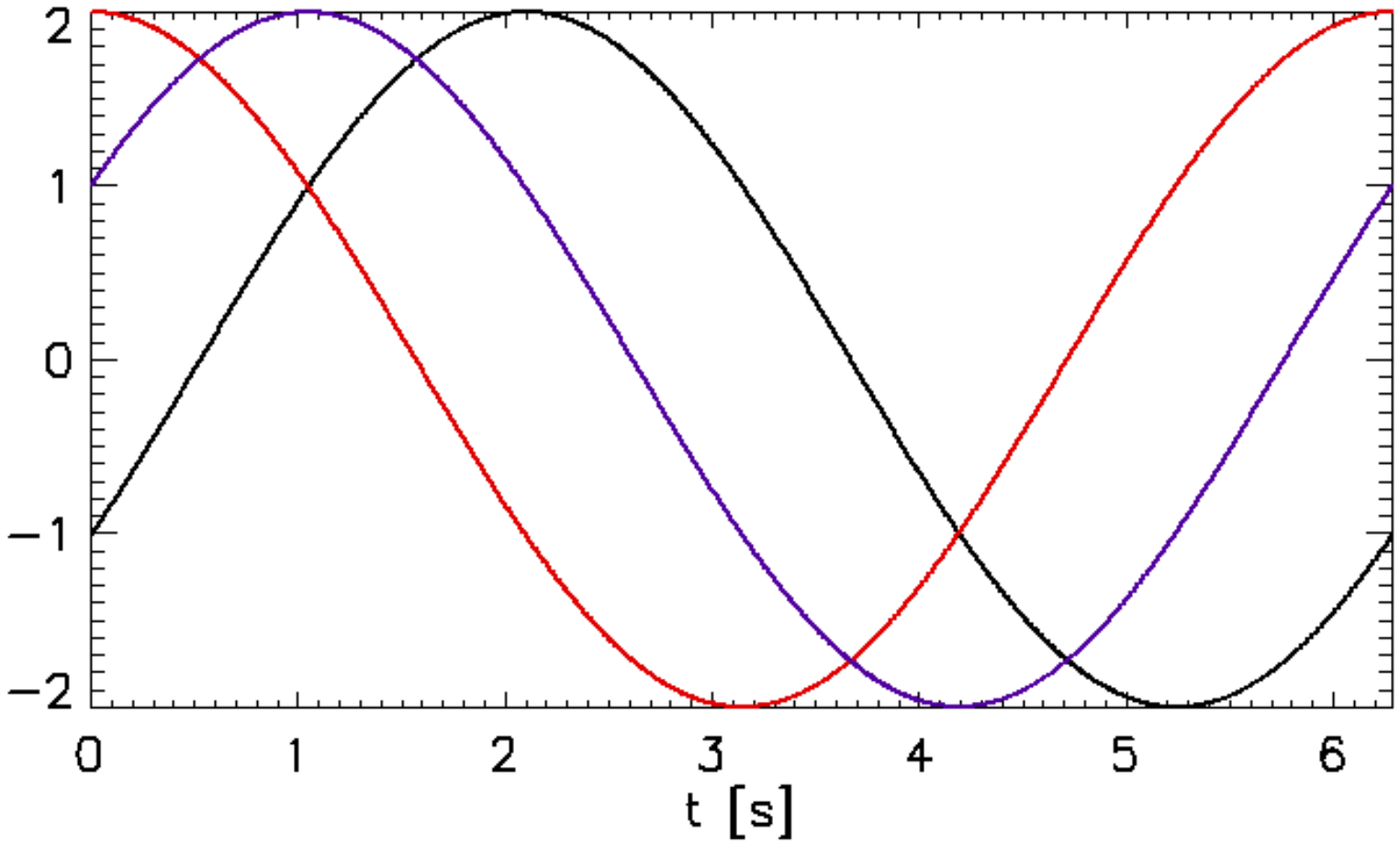
Two waves:

$$y_1(x,t) = (2.0\ \mathrm{m})\sin(kx - \omega t + 7\pi/6)$$
$$y_2(x,t) = (2.0\ \mathrm{m})\cos(kx - \omega t)$$

travel together down a string with ω=1 rad/s.

Find the amplitude and phase of the resulting superposition

$$y_3(x,t) = y_1 + y_2 = A\cos(kx - \omega t + \phi_0)$$



y_1
y_2
$y_3 = y_1 + y_2$

Summary

more info

- The medium does not move at the same speed as the wave. The medium velocity and acceleration are:

$$y = A\cos(kx - \omega t + \phi_0)$$

$$v_y = A\omega\sin(kx - \omega t + \phi_0)$$

$$a_y = -A\omega^2\cos(kx - \omega t + \phi_0)$$

- Wave equation: $\frac{\partial^2 y(x,t)}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y(x,t)}{\partial t^2}$ with $v = \omega/k$
- Waves may be linearly superposed
- Waves can be written in complex notation as $\tilde{y}(x,t) = \tilde{A}e^{j(kx-\omega t)}$

WAVES ON A STRETCHED STRING

EXAMPLE

A 6 m long rope has a mass of 1.0 kg. It is held taut with 12 N of force. How long does it take for a wave to travel from one end to the other?

EXTRA EXAMPLE TO TRY ON YOUR OWN

A mass m is subject to 2 driving forces, which acting separately would produce oscillations of:

$$y_1(t) = (6.00\ \text{cm})\sin(\omega t + \frac{\pi}{4}) \qquad \omega = 127\ \text{rad/s}$$

$$y_2(t) = (4.00\ \text{cm})\sin(\omega t - \frac{\pi}{6})$$

Draw the phasor diagram. Label the phasor for each oscillation, the resultant phasor, and clearly show the relation between the resultant phasor and the 2 individual phasors.

Lecture 8: Sep 22, 2026

Last time:

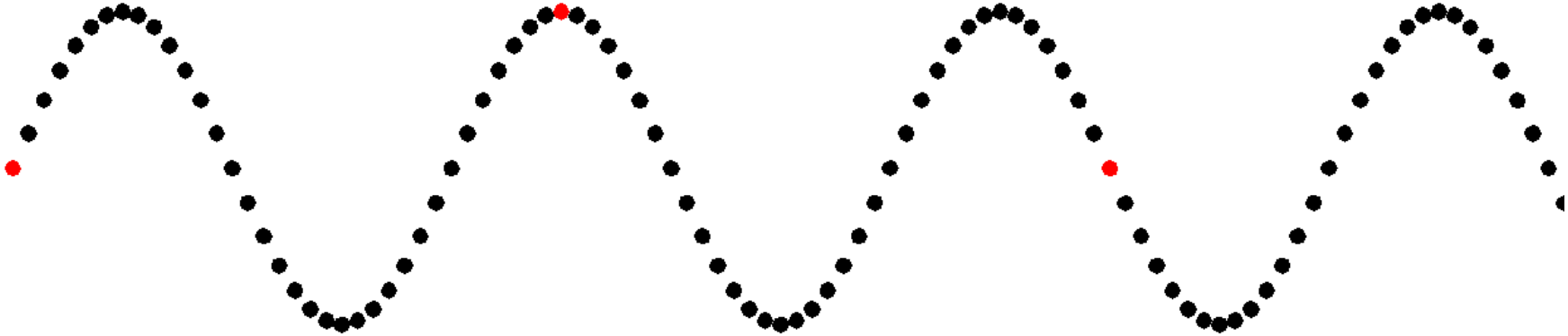
- Complex representation for wave addition (interference)
- Waves on a stretched string
- Using dimensional analysis

Today's class:

- Sound waves in a solid (Young's modulus)
- Sound waves in fluids (bulk modulus)
- Speed of sound in ideal gases

TRANSVERSE VS LONGITUDINAL WAVES

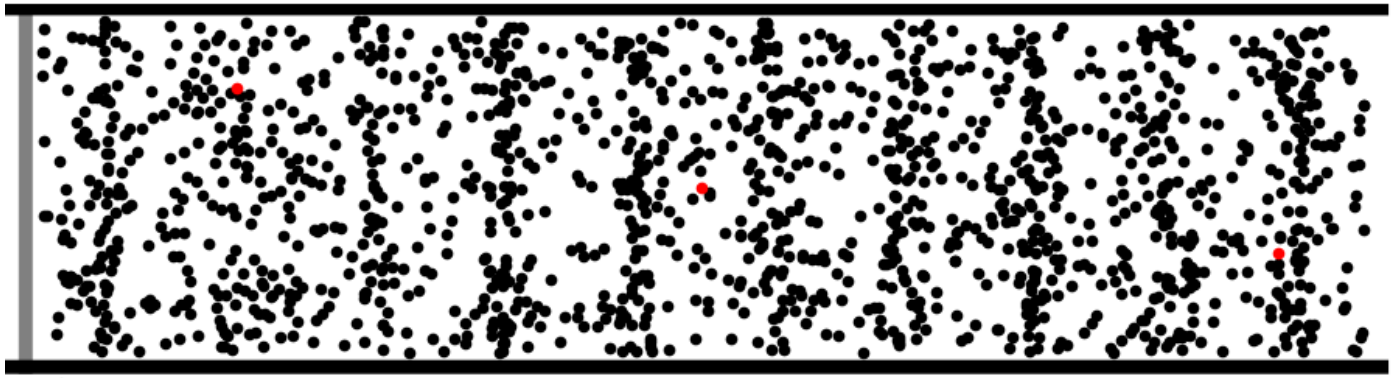
Transverse waves



http://www.phy.olemiss.edu/~perera/animations/waves.html

- Particle displacement is sinusoidal and perpendicular to the wave motion (transverse wave)
- Example is a wave travelling on a string with the "particles" being bits of the string

Longitudinal waves in a medium



Animation: Dr. Dan Russell, Grad. Prog. Acoustics, Penn State:
http://www.acs.psu.edu/drussell/Demos/waves/wavemotion.html

- Particle displacement is sinusoidal and parallel to the wave motion (longitudinal wave)
- Where the particles bunch together, the density and pressure are greater

Numeric Top Hat



When a pile is driven into the ground, the impact is transmitted along the length of the pile as a longitudinal wave. How long does it take for the wave to travel the length of a 40m steel pile?

Typical values for steel:

Y=200 GPa

ρ=7.8x10^3 kg/m^3

Answer in units of ms (but just input a number) to 2 significant digits.

Top Hat Answer



Typical values for steel:

Y=200 GPa

ρ=7.8x10^3 kg/m^3

Wave velocity: $v = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{200\times10^9\ \text{Pa}}{7.8\times10^3\ \text{kg/m}^3}} = 5060\ \text{m/s}$

Travel time: t = 40m/5060 m/s = **7.9 ms**

Example

Speed of sound in ^{4}He at 20°C:

$$v = \sqrt{\frac{\gamma RT}{M}}$$

M=4.00 g/mol=4x10^{-3} kg/mol

(2 neutrons and 2 protons)

T=273+20=293 K

γ=5/3 (monatomic)

v= 1010 m/s

Example

Speed of sound in air (80% N_2, 20% O_2) at 20°C:

$$v = \sqrt{\frac{\gamma RT}{M}}$$

Atomic mass of N=14.0 u

Atomic mass of O=16.0 u

M=0.2(2*16.0g/mol)+0.8(2*14.0g/mol)

= 28.8 g/mol

γ=7/5 (diatomic)

T=273+20=293 K

→v=344 m/s

Summary of wave speeds

Fluids and gases: $v = \sqrt{B/\rho}$

ρ=mass density (kg/m^3)
B=Bulk modulus (Pa)

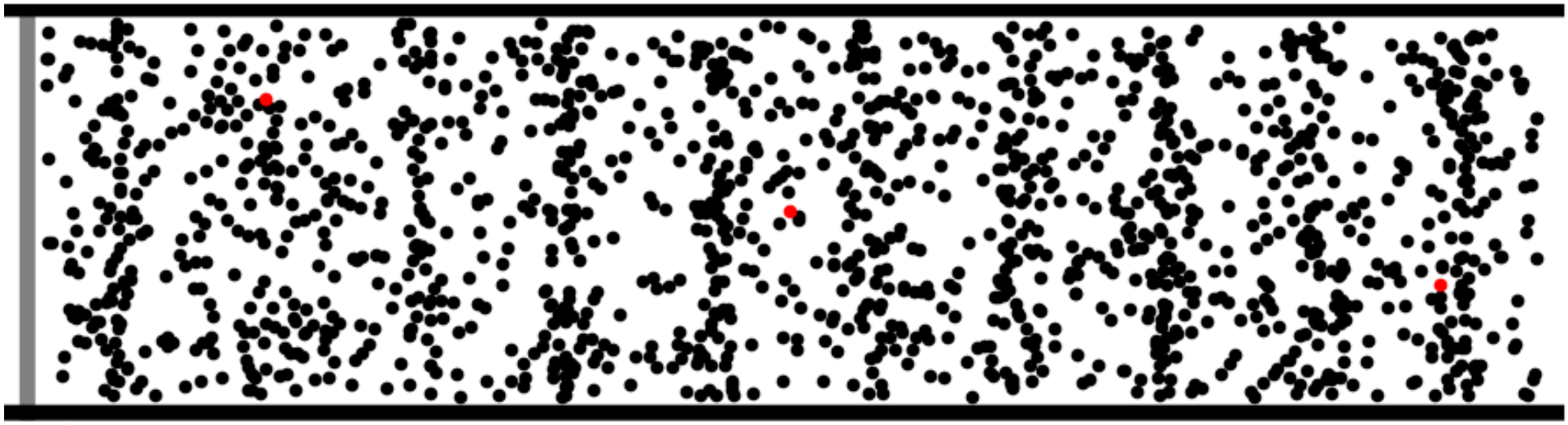
Ideal gases: $v = \sqrt{\gamma P/\rho} = \sqrt{\gamma RT/M}$

P=pressure (Pa)
γ=5/3 (monatomic), 7/5 (diatomic)
R=8.314 J/mol.K
M=mass per mole (kg)

Solid rod: $v = \sqrt{Y/\rho}$

Y=Young's modulus (Pa)

Pressure and displacement out of phase



Animation: Dr. Dan Russell, Grad. Prog. Acoustics, Penn State:
http://www.acs.psu.edu/drussell/Demos/waves/wavemotion.html